Subject Description Form

Subject Code	FSN5033
Subject Title	Clinical Biochemistry and Molecular Nutrition
Credit Value	3
Level	5
Pre-requisite	Nil
Objectives	This subject is intended to provide an integrated understanding of clinical biochemistry and molecular nutrition. After taking the courses, the students would interpret key laboratory indicators and understand how genetic variations and varied gut microbiome compositions contribute to different metabolic responses to different diets and lifestyles. In addition, the subject is intended to introduce the research techniques and methodologies employed in molecular nutrition studies.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) Understand key biochemical pathways and laboratory indicators relevant to human health and disease. b) Interpret clinical biochemical data commonly used in clinical and nutritional settings. c) Understand basic concepts on genetics, epigenetics, gut microbiota, metabolisms and nutrition d) Understand research techniques and methodologies employed in the nutritional epigenetics and gut microbiota studies e) Understand biological mechanisms behind interactions among genetic variations, gut microbiota, diet and health f) Develop analytical, critical thinking, and written communication skills.
Subject Synopsis/ Indicative Syllabus	The subject will contain 3 modules. The first module focuses on clinical biochemistry, introducing fundamental biochemical principles and clinical biochemical assessments relevant to nutrition and metabolic health. The second module will include basic concepts of nutrigenetics, nutrigenomics and interactions between epigenetics, diet and health. The third module will include interactions between gut microbiota, diet and health. Brief contents of each module are listed below.

	The first module:
	This module introduces the biochemical basis of human metabolism, key biochemical pathways and its application in clinical and nutritional assessment.
	• Basic biochemistry and molecular signaling pathways
	• Principles of clinical biochemistry, interpretation and application
	    o Liver function tests and biochemical assessment of liver diseases
	    o Haematological parameters relevant to nutritional status and metabolic disorders
	    o Endocrine regulation of metabolism and biochemical assessment of hormonal function
	    o Clinical biochemical markers in nutrition-related diseases
	
	The second module:
	This module focuses on the genetic basis of nutritional regulation of metabolism.
	• Concept of molecular nutrition
	• Nuclear receptors as nutrient sensors
	• Epigenetic regulation in diet–gene interactions
	• Genetic polymorphisms and metabolic variations upon different diets
	• Genetic polymorphisms, diet and metabolic diseases
	• Relevant research techniques like metabolomics and genome wide association study
	
	The third module:
	This module focuses on interactions between gut microbiota and hosts, gut-brain axis, and related diseases.
	• Gut microbiome and nutrient metabolisms throughout the lifespan
	• Dietary modulations of gut microbiome by prebiotics and cultural food products
	• Gut microbiome, diet, and metabolic diseases, like obesity, fatty liver disease and inflammatory bowel disease
	• Altering gut microbiome for cognitive benefits
	• Related research technique, like metagenomics

Assessment Methods in Alignment with Intended Learning Outcomes	see table below

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
		a	b	c	d	e	f
1. Quiz	30	√	√	√	√	√	√
2. Group presentation	20	√	√	√	√	√	√
3. Final examination	50	√	√	√	√	√	√
Total	100 %						

Student Study Effort Expected

Class contact:	
▪ Lecture	24 hours
▪ Tutorial	12 hours
▪ Group presentation	3 hours
Other student study effort:	
▪ Self study	84 hours
Total student study effort	123 hours

Reading List and References

1. Clinical Chemistry, Marshall, William J.; Lapsley, Márta; Day, Andrew P.; Shipman, Kate E., Amsterdam: Elsevier; 2023.
2. Tietz Textbook of Clinical Chemistry and Molecular Diagnostics, Burtis, Carl A.; Ashwood, Edward R.; Bruns, David E., Amsterdam: Elsevier; 2018.
3. Nutritional Epigenomics, Ferguson, Bradley S, San Diego: Elsevier Science & Technology; 2019; 1
4. Nutrigenetics applying the science of personal nutrition, Kohlmeier, Martin ; Kohlmeier, Gabrielle Z. ; ScienceDirect (Online service)
5. Nutriomics: Well-being through Nutrition, Thangadurai, Devarajan ; Islam, Saher ; Nollet, Leo M. L ; Adetunji, Juliana ; Nollet, Leo M.L ; Islam, Saher ; Thangadurai , Devarajan ; Islam, Saher ; Thangadurai, Devarajan ; Nollet, Leo M.L. ; Adetunji, Juliana Bunmi, United States: CRC Press; 2022; 1
6. Nutrigenomics : how science works, Carlberg, Carsten ; Ulven, Stine Marie. ; Molnár, Ferdinand. Cham : Springer; 2020
7. Gene Polymorphism and Nutrition: Relationships with Chronic Disease, Crujeiras, Ana B ; de Luis Roman, Daniel-

	Antonio, Basel: MDPI - Multidisciplinary Digital Publishing Institute; 2023 8. Dietary Influence on Nutritional Epidemiology, Public Health and Our Lifestyle, Varela, Lourdes M, Basel: MDPI - Multidisciplinary Digital Publishing Institute; 2023 9. Gut Microbiota: Interactive Effects on Nutrition and Health, Ishiguro, Edward ; Haskey, Natasha ; Campbell, Kristina, San Diego: Elsevier Science & Technology; 2023; Second edition 10. Metabolomics and Gut Microbiota in Nutrition and Disease, Kochhar, Sunil ; Martin, François-Pierre ; Martin, François-Pierre; Kochhar, Sunil, London: Springer London, Limited; 2014; 2015 11. The Gut-Brain Axis: Dietary, Probiotic, and Prebiotic Interventions on the Microbiota, Hyland, Niall ; Stanton, Catherine, San Diego: Elsevier Science & Technology; 2023; Second edition 12. The Gut Microbiome: Bench to Table, Wu, Vivian C. H ; Wu, Vivian C.H ; Wu, Vivian C.H. United States: CRC Press; 2022; 1